

Molecular Docking Analysis of Catechin Gallate with SARS-CoV-2 Main Protease (NSP5) Using SwissDock and AutoDock Vina

Protein Name : SARS-CoV-2 Main Protease (NSP5) – PDB ID: 6LU7

Ligand Name: Catechin Gallate

Docking Workflow :

1. PyMOL

- Preprocessed the protein structure
- Water molecules removed
- Hydrogen atoms added to protein

2. AutoDock Vina

- Ligand preparation
- Gasteiger charges added

3. SwissDock

- Performed the docking
- Best binding pose selected for analysis

Grid Box / Binding Site Parameters (SwissDock) :

Center coordinates (x, y, z) : -11.824, 14.735, 74.152

Box size (Å) (x, y, z): 25, 25, 25